SEQUENTIAL REALIZABILITY OF A FLAT VERTEX: A RIGID LAYERED MODEL AND ONE SIX-CREASE CONFIGURATION

Abstract. We fix a combinatorial model of folding a single flat vertex by a sequence of steps, each of which folds along one line of the current folded image and may create and erase creases. We add a layered refinement in which each step is a rigid half-turn of a block of layers. For the vertex $A = \{0, 1, 3, 4, 5, 7\} \cdot \pi/4$ we show that a natural history with cancellation, namely *fold both diagonals, fold the full cross, unfold the line* $\{\pi/2, 3\pi/2\}$, is valid in the base model and passes the collision test (C), but admits no rigid realization. We introduce the *surviving partition* of a history, which groups the final creases by the step that last created them and forgets the order of the steps. We conjecture that no rigidly realizable history of A has the surviving partition $\{\{0, 4\}, \{1, 5\}, \{3, 7\}\}$ (three straight lines), nor indeed any surviving partition into straight lines through the vertex. We then add reverse folds: each moving piece is given the side over which it is lifted, and the two halves of a flap may be lifted over opposite sides. With the complementary labeling this also describes inside reverse folds. The history above remains unrealizable with reverse folds, and we give a vertex with four creases whose natural history with cancellation is realizable with a reverse fold but not rigidly.

1. Base model

Conventions. $\mathbb{T} := \mathbb{R}/2\pi\mathbb{Z}$ (directions), ordered via representatives in $[0, 2\pi)$; $\mathbb{P} := \mathbb{R}/\pi\mathbb{Z}$ (lines) with projection $q\colon \mathbb{T} \to \mathbb{P}$. (a, b) denotes the open counterclockwise arc from a to b, and cl its closure. For $c \in \mathbb{T}$ let $\sigma_c(x) := 2c - x$. Since $2(c + \pi) = 2c$ in $\mathbb{T}$, σ_c depends only on $q(c)$, so σ_ℓ is defined for $\ell \in \mathbb{P}$. $\mathrm{Iso}(\mathbb{T}) := \{x \mapsto \varepsilon x + t : \varepsilon = \pm 1,\ t \in \mathbb{T}\}$.

Fact 1.1. For $g \in \mathrm{Iso}(\mathbb{T})$ and $x \in \mathbb{T}$: $g \circ \sigma_x \circ g^{-1} = \sigma_{g(x)}$.

Proof. For $g(y) = \varepsilon y + t$: $g\sigma_x g^{-1}(y) = \varepsilon\big(2x - \varepsilon(y - t)\big) + t = 2(\varepsilon x + t) - y$. □

Definition 1.2 (Faces, folding maps, flatness)**.** Let $S \subset \mathbb{T}$ be finite, $S = \{s_1 < \cdots < s_m\}$. If $m \geq 1$ the *faces* of S are $F_j := (s_j, s_{j+1})$ (indices mod m); if $S = \emptyset$ the only face is $\mathbb{T}$. The *folding maps* are $\varphi_1 := \mathrm{id}$ and $\varphi_{j+1} := \varphi_j \circ \sigma_{s_{j+1}}$ for $1 \leq j < m$. S is *flat* if $S = \emptyset$ or $\varphi_m \circ \sigma_{s_1} = \mathrm{id}$.

Fact 1.3. Let $S \neq \emptyset$ be flat. Then m is even and every face has length $\leq \pi$ (Kawasaki). The map $\Phi_S\colon \mathbb{T} \to \mathbb{T}$, $\Phi_S|_{\mathrm{cl}\, F_j} := \varphi_j$, is well defined and continuous, since $\varphi_{j+1}(s_{j+1}) = \varphi_j\sigma_{s_{j+1}}(s_{j+1}) = \varphi_j(s_{j+1})$ and, at s_1, $\varphi_m(s_1) = \varphi_m\sigma_{s_1}(s_1) = s_1 = \varphi_1(s_1)$. Set $\Phi_\emptyset := \mathrm{id}$.

Definition 1.4. For flat S and $\ell \in \mathbb{P}$ let $\Pi_S(\ell) := \{x \in \mathbb{T} : q(\Phi_S(x)) = \ell\}$. It is finite: Φ_S is an isometry on each closed face, which has length $\leq \pi$, so each closed face contains at most two points of $\Pi_S(\ell)$; and $\Pi_\emptyset(\ell) = q^{-1}(\ell)$.

Definition 1.5 (Step)**.** A *step* from a flat S is a nonempty $D \subset \mathbb{T}$ with $|D|$ even and $D \subseteq \Pi_S(\ell)$ for some $\ell \in \mathbb{P}$; this ℓ is unique and denoted $\ell(D)$. Put $N(D) := D \setminus S$ (created), $K(D) := D \cap S$ (erased). The result is $S \triangle D$. This is exactly the original definition: (P) is $|D|$ even, and (L) is $D \subseteq \Pi_S(\ell)$.

Definition 1.6 (Pieces and labelings)**.** The *pieces* of a step D from S are the faces of $S \cup D$. A *labeling* is a map $\mu\colon \{\text{pieces}\} \to \{0, 1\}$ such that two consecutive pieces receive different labels iff their common endpoint lies in D. As $|D|$ is even there are exactly two labelings, and they

are complementary. Label 1 means *moving*. The parent face of a piece p, i.e. the face of S containing p, is denoted $\pi(p)$.

Lemma 1.7. *Let D be a step from flat S, $\ell = \ell(D)$, and μ a labeling. Define $\Psi_\mu := \Phi_S$ on closures of pieces with $\mu = 0$ and $\Psi_\mu := \sigma_\ell \circ \Phi_S$ on closures of pieces with $\mu = 1$. Then $S \triangle D$ is flat, and $\Phi_{S\triangle D} = g \circ \Psi_\mu$ for some $g \in \mathrm{Iso}(\mathbb{T})$.*

Proof. Ψ_μ is well defined and continuous because σ_ℓ fixes $\Phi_S(D) \subseteq q^{-1}(\ell)$ pointwise. Compare the restrictions of Ψ_μ to the two pieces meeting at a point $x \in S \cup D$. Below we treat the case where the piece before x has label 0; the other case follows by composing both restrictions on the left with σ_ℓ. If $S = \emptyset$, read $\varphi_j = \mathrm{id}$.

- If $x \in S \setminus D$, $x = s_{j+1}$, the labels agree and the restrictions are φ_j and $\varphi_{j+1} = \varphi_j \sigma_x$.
- If $x \in N(D)$ lies in a face F_j, then $q(\varphi_j(x)) = \ell$, so by Fact 1.1 $\varphi_j \sigma_x = \sigma_{\varphi_j(x)} \varphi_j = \sigma_\ell \varphi_j$. So the restrictions are φ_j and $\sigma_\ell \varphi_j = \varphi_j \sigma_x$, and x becomes a fold.
- If $x \in K(D)$, $x = s_{j+1}$, then the restrictions are φ_j and $\sigma_\ell \varphi_{j+1} = \sigma_\ell \varphi_j \sigma_x = \sigma_\ell \sigma_\ell \varphi_j = \varphi_j$, using $\varphi_j \sigma_x = \sigma_\ell \varphi_j$ as before. So they coincide and the fold at x disappears.

Thus Ψ_μ is continuous, piecewise in $\mathrm{Iso}(\mathbb{T})$, and changes by σ_x exactly at the points x of $S \triangle D$. Let $g \in \mathrm{Iso}(\mathbb{T})$ make $g \circ \Psi_\mu$ the identity on the first face of $S \triangle D$. Then the folding maps of $S \triangle D$ are the restrictions of $g \circ \Psi_\mu$, and the closing condition holds because $g \circ \Psi_\mu$ is continuous around the whole circle. (If $S \triangle D = \emptyset$, Ψ_μ is a single isometry and $g = \Psi_\mu^{-1}$.) □

Lemma 1.8 (Reversibility)**.** *If D is a step from S, then D is a step from $S \triangle D$.*

Proof. For $d \in D$, $\Psi_\mu(d) = \Phi_S(d)$ because σ_ℓ fixes $q^{-1}(\ell)$ pointwise. By Lemma 1.7, $\Phi_{S\triangle D}(d) = g(\Phi_S(d)) \in g\big(q^{-1}(\ell)\big)$, a single line, for all $d \in D$. □

Definition 1.9 (Histories)**.** A *history* is a tuple $H = (D_1, \dots, D_n)$ with $S_0 = \emptyset$, $S_i = S_{i-1} \triangle D_i$, and each D_i a step from S_{i-1}. It is a history of A if $S_n = A$. For $a \in A$ let $\tau_H(a) := \max\{i : a \in N(D_i)\}$; the set is nonempty since $a \notin S_0$. The *structure* $\mathrm{struct}(H)$ is the preorder $a \preceq b \iff \tau_H(a) \le \tau_H(b)$. The *cost* is $\mathrm{cost}(H) := \sum_i |K(D_i)|$. $R(A)$ is the set of structures of histories of A, and $R_0(A)$ the subset coming from histories of cost 0.

Definition 1.10 (Surviving partition, straight blocks)**.** The *surviving partition* of a history $H = (D_1, \dots, D_n)$ of A is

$$\mathcal{P}(H) := \big\{\tau_H^{-1}(i) : 1 \le i \le n\big\} \setminus \{\emptyset\},$$

a partition of A. Its blocks are called the *surviving blocks* of H: the block $\tau_H^{-1}(i)$ is what survives in A of the creations made by step i. A set $B \subseteq \mathbb{T}$ is *straight* if $B + \pi = B$, i.e. B is a union of lines through the vertex.

Remark 1.11. $\mathcal{P}(H)$ is the partition underlying $\mathrm{struct}(H)$, with the order of the classes forgotten. It ignores scaffold creases, i.e. creases created and later erased, and it ignores earlier creations of creases of A that were erased and created again: only the last creation counts.

Definition 1.12 (Collision condition (C))**.** A step D from S satisfies (C) if the following holds. Whenever F, F' are faces of S with $F \cap D = F' \cap D = \emptyset$, and $x \in F$, $x' \in F'$ satisfy $\Phi_S(x) = \Phi_S(x') \in q^{-1}(\ell(D))$, then $\mu(F) = \mu(F')$. Such faces are pieces, and the condition does not depend on which labeling is used.

2. Layered model

Definition 2.1 (Layered state)**.** Let S be flat. For a face F put $W_F := \Phi_S(F)$, an open arc (or $\mathbb{T}$ if $S = \emptyset$). Let $E_S := \Phi_S(S)$, a finite set, and for $u \notin E_S$ let $L(u) := \{F : u \in W_F\}$. A *layer order* on S is a family $(<_u)_{u \in \mathbb{T} \setminus E_S}$ of strict total orders on $L(u)$ that is locally constant in u.

For $v \in E_S$, write $<_{v^+}$ and $<_{v^-}$ for the order at points of $\mathbb{T}\setminus E_S$ immediately counterclockwise, resp. clockwise, of v; likewise $L(v^\pm)$. Put $\Theta(v) := L(v^-) \cap L(v^+)$, the faces *passing through* v. The *taco* of a ray $s = s_{j+1} \in S$ is $\{F_j, F_{j+1}\}$. Since $\varphi_{j+1}(s+t) = \varphi_j(s-t)$, both faces of the taco of s lie in $L(v^\epsilon)$ for exactly one sign $\epsilon = \epsilon(s)$, where $v = \Phi_S(s)$. The pair $(S, <)$ is a *layered state* if the following three conditions hold.

(O1) For every $v \in E_S$, the orders $<_{v^-}$ and $<_{v^+}$ agree on $\Theta(v)$.
(O2) If $s \neq s'$, $\Phi_S(s) = \Phi_S(s') = v$ and $\epsilon(s) = \epsilon(s') = \epsilon$, then the two tacos, which are disjoint, do not interleave in $<_{v^\epsilon}$: it is not the case that exactly one face of one taco lies strictly between the two faces of the other.
(O3) If $v = \Phi_S(s)$ and $c \in \Theta(v)$, then c does not lie strictly between the two faces of the taco of s in $<_{v^{\epsilon(s)}}$.

The *initial layered state* is $\emptyset$ with its unique layer order.

Definition 2.2 (Rigid step)**.** The data are a layered state $(S, <)$, a step D with $\ell = \ell(D)$, a labeling μ, and a choice of one component $H^\uparrow$ of $\mathbb{T} \setminus q^{-1}(\ell)$; the other component is $H^\downarrow$. Physically: a half-turn about ℓ that lifts the layers over $H^\uparrow$ and lowers those over $H^\downarrow$.

(F) Let $u \notin E_S \cup q^{-1}(\ell)$, and let p, p' be pieces with $\mu(p) = 1$, $\mu(p') = 0$ and $u \in \Phi_S(p) \cap \Phi_S(p')$. Then $\pi(p') <_u \pi(p)$ if $u \in H^\uparrow$, and $\pi(p) <_u \pi(p')$ if $u \in H^\downarrow$.

New order. Let $S' = S \triangle D$ and let $\Psi = \Psi_\mu$ and g be as in Lemma 1.7. Each face G of S' is a union of pieces and points of $K(D)$, and Ψ is injective on G. Let u avoid the finite set $E_S \cup \sigma_\ell(E_S) \cup q^{-1}(\ell)$. Define $\prec_u$ on the pieces p with $u \in \Psi(p)$ as follows:

(i) two staying pieces: $p \prec_u p' \iff \pi(p) <_u \pi(p')$;
(ii) two moving pieces: $p \prec_u p' \iff \pi(p') <_{\sigma_\ell(u)} \pi(p)$, i.e. the order is reversed;
(iii) p moving and p' staying: $p' \prec_u p$ if $u \in H^\downarrow$, and $p \prec_u p'$ if $u \in H^\uparrow$.

Set $G <'_{g(u)} G'$ iff $p_G \prec_u p_{G'}$, where p_G is the unique piece of G with $u \in \Psi(p_G)$. The step is *rigid* if (F) holds and $<'$ extends to a layer order making $(S', <')$ a layered state.

A history is *rigidly realizable* if there are layered states $(S_i, <^i)$, starting from the initial one, such that each transition $(S_{i-1}, <^{i-1}) \to (S_i, <^i)$ is a rigid step for D_i, for some labeling and some choice of $H^\uparrow$. Let

$$\mathcal{P}_{\mathrm{rig}}(A) := \{\mathcal{P}(H) : H \text{ a rigidly realizable history of } A\}$$

be the set of *rigidly realizable surviving partitions* of A.

Lemma 2.3. *(F) implies (C). Consequently every step of a rigidly realizable history satisfies (C).*

Proof. Let F, F', x, x' be as in Definition 1.12 with $w := \Phi_S(x) = \Phi_S(x')$, and suppose $\mu(F) = 1 \neq 0 = \mu(F')$. Since x, x' are interior points of faces, $F, F' \in \Theta(w)$ if $w \in E_S$, and otherwise $F, F' \in L(u)$ for all u near w on both sides; in both cases their relative order is the same on both sides of w (by (O1), resp. local constancy). But (F) requires $\pi(F') <_u \pi(F)$ for $u \in H^\uparrow$ near w and $\pi(F) <_u \pi(F')$ for $u \in H^\downarrow$ near w. Contradiction. □

Lemma 2.4 (Simple folds)**.** *Let $D = \Pi_S(\ell)$ with $D \cap S = \emptyset$. Then the image of each piece lies in one component of $\mathbb{T} \setminus q^{-1}(\ell)$, and each labeling moves exactly the pieces lying over one component. Hence (F) holds vacuously, and for each of the four choices of labeling and $H^\uparrow$ the order $<'$ is determined by (i)–(iii).*

Proof. No ray maps into $q^{-1}(\ell)$, since $\Pi_S(\ell) \cap S = D \cap S = \emptyset$, and all interior points of faces mapping into $q^{-1}(\ell)$ lie in D. So no piece's image meets $q^{-1}(\ell)$ in its interior. Across a point

of D (a transversal crossing) both the side and the label switch; across a ray of S (a fold away from ℓ) neither switches. □

3. Reverse folds

In a rigid step every moving piece turns the same way. In a reverse fold the two halves of a flap turn in opposite senses: the fold of the flap opens during the move and closes again with its two faces exchanged. We record this by a sense for each moving piece.

Definition 3.1 (Runs, joints, senses)**.** Let D be a step from a flat S, with a labeling μ. A *moving run* is a maximal sequence $p_1, \dots, p_r$ of moving pieces that are consecutive counterclockwise, p_i and p_{i+1} sharing an endpoint; the shared endpoints lie in $S \setminus D$ and are the *joints* of the run. Fix a component $H^{\uparrow}$ of $\mathbb{T} \setminus q^{-1}(\ell)$ as in Definition 2.2. A *sense assignment* is a map τ from the moving pieces to $\{0,1\}$ such that on each moving run either $\tau \equiv 0$, or there is exactly one joint s, the *spine* of the run, such that τ is constant on the pieces of the run before s, constant on those after s, and different on the two sides. Put $U(p) := H^{\uparrow}$ if $\tau(p) = 0$ and $U(p) := H^{\downarrow}$ if $\tau(p) = 1$.

Definition 3.2 (Step with senses)**.** The data are those of Definition 2.2 together with a sense assignment τ.

(F′) Let $u \notin E_S \cup q^{-1}(\ell)$, and let p be a moving and p' a staying piece with $u \in \Phi_S(p) \cap \Phi_S(p')$. Then $\pi(p') <_u \pi(p)$ if $u \in U(p)$, and $\pi(p) <_u \pi(p')$ if $u \notin U(p)$.

(F″) Let u be as in (F′), and let p, p' be moving pieces with $\tau(p) \neq \tau(p')$ and $u \in \Phi_S(p) \cap \Phi_S(p')$. If $u \in U(p)$, then $\pi(p') <_u \pi(p)$.

New order. Let S', Ψ, g and u be as in Definition 2.2. Split the pieces p with $u \in \Psi(p)$ into three sets: B_u, the moving pieces with $u \in U(p)$; C_u, the staying pieces; and A_u, the moving pieces with $u \notin U(p)$. Define $\prec_u$ by $B_u \prec_u C_u \prec_u A_u$, by rule (i) of Definition 2.2 on C_u, and by rule (ii) on B_u and on A_u. Set $G <'_{g(u)} G'$ iff $p_G \prec_u p_{G'}$, as in Definition 2.2.

The step is *rigid with senses* if (F′) and (F″) hold and $<'$ extends to a layer order making $(S', <')$ a layered state. A history is *realizable with reverse folds* if there are layered states $(S_i, <^i)$, starting from the initial one, such that each transition is a rigid step with senses for D_i, for some labeling, choice of $H^{\uparrow}$ and sense assignment. Let $\mathcal{P}_{\mathrm{rev}}(A)$ be the set of surviving partitions of such histories of A.

Lemma 3.3.

(1) $\prec_u$ is a strict total order, and all pieces of B_u, as well as all pieces of A_u, have the same sense.
(2) If $\tau \equiv 0$, a step with senses is a rigid step in the sense of Definition 2.2. Hence $\mathcal{P}_{\mathrm{rig}}(A) \subseteq \mathcal{P}_{\mathrm{rev}}(A)$.
(3) (F′) implies (C). Consequently every step of a history realizable with reverse folds satisfies (C).

Proof. (1) $U(p)$ is one of the disjoint sets $H^{\uparrow}$, $H^{\downarrow}$. All $p \in B_u$ have $u \in U(p)$, so $U(p)$ is the component containing u and $\tau(p)$ is the same for all of them; likewise for A_u. Two staying pieces with the same parent have disjoint images under Φ_S, and two moving pieces with the same parent have disjoint images under $\Psi = \sigma_\ell \circ \Phi_S$; so within C_u, and within B_u and A_u, distinct pieces have distinct parents. These parents lie in $L(u)$ for C_u and in $L(\sigma_\ell(u))$ for B_u and A_u, since $u \in \Psi(p)$ means $\sigma_\ell(u) \in \Phi_S(p)$ for a moving p. Rules (i) and (ii) therefore order each set totally.

(2) If $\tau \equiv 0$, then $U(p) = H^{\uparrow}$ for every moving p, (F′) is (F), and (F″) is vacuous. For $u \in H^{\uparrow}$ the set A_u is empty and the moving pieces over u lie below the staying ones; for $u \in H^{\downarrow}$ the set B_u is empty and they lie above. This is rule (iii).

(3) As in the proof of Lemma 2.3, with $H^{\uparrow}$ replaced by $U(F)$: near $w \in q^{-1}(\ell)$ the points on one side of w lie in $U(F)$ and those on the other side do not, so (F′) requires opposite relative orders of F and F' on the two sides of w, while the relative order is the same on both sides. □

Remark 3.4 (Reading). Each moving piece p is turned about ℓ by a half-turn that lifts it over $U(p)$. Pieces of one sense turn together, as in a rigid step. Across a spine the two halves of a run turn in opposite senses. (F′) says that every moving piece leaves without passing through staying paper, and (F″) that pieces turning in opposite senses move apart instead of through each other. The new order puts the pieces that arrive over the top above the staying ones and those that arrive from underneath below them, each group reversed as in a rigid step.

When the two halves of a split run end up below and above the staying layers around them, the step is an *outside reverse fold* of the flap formed by the run. The two labelings describe the same relative motion from the two sides. With the labeling in which the split run is the base, the base's halves turn around the tip, which ends up between them: an *inside reverse fold* of the tip.

With the staying paper fixed, opposite turns are an idealization. The two halves of a run are hinged to staying paper along rays whose images lie on ℓ, so both hinges are turns about ℓ, and two faces turned about ℓ by different angles cannot share a spine, which does not map into ℓ. In practice the flap is opened slightly during a reverse fold. The definition records the combinatorics of the move, not a rigid motion of faces.

Example 3.5 (Inside and outside reverse folds)**.** In this example directions are in degrees. Fold in half, $D_1 = \{0, 180\}$, and then fold along a second line, $D_2 = \{\alpha, 360 - \alpha\}$. Each of the two runs of D_2 has one joint: the *tip*, through 0, and the *base*, through 180. Faces are named by their endpoints, and stacks are listed from bottom to top over the arc, in the coordinates of the folding map, where all four faces overlap. The following was computed by exhausting all choices (Appendix B).

- $\alpha = 45$. Rigid steps reach four layered states, and steps with senses two more: $(180, 315) < (315, 360) < (0, 45) < (45, 180)$ and $(45, 180) < (0, 45) < (315, 360) < (180, 315)$. In both, the tip lies between the two halves of the base: inside reverse folds. Both arise only from sense assignments that split the base. Every assignment that splits the tip fails (F″) or (O3): an outside reverse fold would wrap the tip around base layers that pass through the image of its fold.
- $\alpha = 120$. Again four layered states are reachable by rigid steps and two more with senses: $(240, 360) < (180, 240) < (120, 180) < (0, 120)$ and $(0, 120) < (120, 180) < (180, 240) < (240, 360)$. In both, the halves of the tip enclose the base: outside reverse folds. They arise only from assignments that split the tip; every assignment that splits the base fails (F″) or (O3).

Proposition 3.6. *Let $A' = \{3, 8, 10, 13\} \cdot \pi/8$, and let $K = (\{8, 0\}, \{3, 13\}, \{10, 0\})$ in units of $\pi/8$, where 0 is a scaffold crease, created by the first step and erased by the third. Then K is a history of A', every step of K satisfies (C), and:*

(a) K is not rigidly realizable;
(b) K is realizable with reverse folds: rigid steps for D_1 and D_2, and an outside reverse fold for D_3 that splits the run from 10 to 16 at the joint 13.

Proof. All claims were computed (Appendix B); we describe what the computation finds. For (a): after D_1 and D_2, over the arc where the four faces $(0, 3)$, $(3, 8)$, $(8, 13)$, $(13, 16)$ overlap,

every choice of labelings and up sides gives one of four stacks:

$$(0,3) < (13,16) < (8,13) < (3,8), \quad (13,16) < (0,3) < (3,8) < (8,13),$$
$$(3,8) < (8,13) < (13,16) < (0,3), \quad (8,13) < (3,8) < (0,3) < (13,16).$$

Over that arc, the moving pieces of D_3 belong to the faces $(8,13)$ and $(13,16)$ for one labeling, and to $(0,3)$ and $(3,8)$ for the other. In each stack these two faces are either the two middle layers or the bottom and the top layer. So the moving pieces over that arc are neither all above nor all below the staying ones, and (F) fails. For (b), starting from the second stack, the pieces in $(8,13)$ and $(13,16)$ turn in opposite senses: the first, on top, is lifted, and the second, at the bottom, goes underneath. The resulting stack over the arc $(6,8)$ is $(13,3) < (3,8) < (8,10) < (10,13)$, where $(13,3)$ is the face formed by $(13,16)$ and $(0,3)$ once the crease 0 is erased. □

At A' the crease 8 has no continuation: the scaffold crease 0 continues it for two steps, and the reverse fold erases it.

4. The configuration $\{0,1,3,4,5,7\} \cdot \pi/4$

From now on the unit is $\pi/4$, so $\mathbb{T} = \mathbb{R}/8\mathbb{Z}$, and

$$A = \{0,1,3,4,5,7\}.$$

Let $f_k := (k-1,k)$, $k = 1,\dots,8$, be the faces of $\{0,\dots,7\}$. In the drawing convention where direction 0 points up, $\{0,4\}$ is the vertical and $\{2,6\}$ the horizontal line. The three lines of A are $\{0,4\}$, $\{1,5\}$ and $\{3,7\}$, and the straight subsets of A are exactly the unions of these. Consider

$$H_1 = \big(\{0,4\},\ \{1,7\},\ \{3,5\}\big),$$
$$H_2 = \big(\{1,5\},\ \{3,7\},\ \{0,2,4,6\},\ \{2,6\}\big),$$
$$H_3 = \big(\{1,5\},\ \{3,7\},\ \{0,4\}\big).$$

Proposition 4.1.

(a) H_1, H_2, H_3 *are histories of* A.
(b) $\mathcal{P}(H_1) = \{\{0,4\},\{1,7\},\{3,5\}\}$*: one straight block and two non-straight blocks, which are mirror images of each other across the line* $\{0,4\}$ *(since* $7 = -1$, $5 = -3$*).* $\mathcal{P}(H_2) = \mathcal{P}(H_3) = \Lambda := \{\{0,4\},\{1,5\},\{3,7\}\}$*: three straight blocks.*
(c) $\mathrm{cost}(H_1) = \mathrm{cost}(H_3) = 0$ *and* $\mathrm{cost}(H_2) = 2$*. In particular* Λ *is the surviving partition of a history of cost* 0*, namely* H_3.
(d) Every step of H_1 *and* H_2 *satisfies (C). Step* 3 *of* H_3 *violates (C).*

Proof. We record the normalised folding maps (Definition 1.2) face by face.

H_1. $\Pi_\emptyset(q(0)) = \{0,4\} = D_1$. For $S_1 = \{0,4\}$: $\Phi = x$ on $(0,4)$ and $-x$ on $(4,8)$, so $\Pi_{S_1}(q(1)) = \{1,7\} = D_2$. For $S_2 = \{0,1,4,7\}$, Φ is x, $2-x$, $x+2$, $-x$ on $(0,1),(1,4),(4,7),(7,8)$ respectively, so $\Phi(3) = \Phi(5) = 7$ and $\Pi_{S_2}(q(7)) = \{3,5\} = D_3$.

H_2, H_3. $\Pi_\emptyset(q(1)) = \{1,5\} = D_1$. For $S_1 = \{1,5\}$: $\Phi = x$ on $(1,5)$ and $2-x$ on $(5,9)$, so $\Pi_{S_1}(q(3)) = \{3,7\} = D_2$. For $S_2 = \{1,3,5,7\}$: $\Phi = x$, $6-x$, $x-4$, $2-x$ on $(1,3),(3,5),(5,7),(7,9)$; rays map to $1,3,1,3$, and $\Phi(0) = \Phi(2) = \Phi(4) = \Phi(6) = 2$, so $\Pi_{S_2}(q(2)) = \{0,2,4,6\}$. Hence D_3 is a step for H_2, and so is $\{0,4\}$ for H_3. For $S_3 = \{0,\dots,7\}$ in H_2: $\Phi(k) = 0$ for even k and $\Phi(k) = 1$ for odd k, with all faces mapping onto $(0,1)$, so $\Pi_{S_3}(q(0)) = \{0,2,4,6\} \supseteq D_4$.

(b) and (c) are read off directly. For (d): in H_1 and in steps 1–3 of H_2 we have $D = \Pi_S(\ell)$, so no face avoiding D contains a preimage of ℓ. In step 4 of H_2, $\Pi_{S_3}(q(0))$ consists of rays, so no face contains one in its interior. In step 3 of H_3, the faces $(1,3)$ and $(5,7)$ of S_2 avoid $\{0,4\}$

and contain 2 and 6, with $\Phi(2) = \Phi(6) = 2 \in q^{-1}(q(2))$; the labeling switches at 0 and 4, so these faces receive opposite labels. □

Theorem 4.2. *H_2 is not rigidly realizable.*

Proof. Steps 1–3 of H_2 satisfy the hypothesis of Lemma 2.4, so each admits exactly four candidate realizations (two labelings times two choices of $H^{\uparrow}$), each determined by (i)–(iii). Running all $4^3 = 64$ combinations from the initial layered state (Appendix A) yields, after step 3, eight faces $f_1, \dots, f_8$ that all have the same image arc W (up to the global isometry g), of length 1 with an endpoint in $q^{-1}(\ell_4)$. Their order $<_W$, bottom to top, is one of the following eight; the second column is the reversal of the first.

$$\begin{array}{ll} f_2f_1f_6f_5\, f_4f_7f_8f_3 & f_3f_8f_7f_4\, f_5f_6f_1f_2 \\ f_4f_7f_8f_3\, f_2f_1f_6f_5 & f_5f_6f_1f_2\, f_3f_8f_7f_4 \\ f_6f_5f_2f_1\, f_8f_3f_4f_7 & f_7f_4f_3f_8\, f_1f_2f_5f_6 \\ f_8f_3f_4f_7\, f_6f_5f_2f_1 & f_1f_2f_5f_6\, f_7f_4f_3f_8 \end{array}$$

In step 4 ($D_4 = \{2, 6\}$) the two labelings move $M = \{f_3, f_4, f_5, f_6\}$ or its complement. Since W lies in a single component of $\mathbb{T} \setminus q^{-1}(\ell_4)$, condition (F) at any $u \in W$ says that the moving set is an up-set of $<_W$ (if $W \subseteq H^{\uparrow}$) or a down-set (if $W \subseteq H^{\downarrow}$). In each of the eight orders, neither M nor its complement is the set of the four lowest faces. Hence (F) fails for every choice. □

Remark 4.3. Theorem 4.2 does not depend on whether simple folds preserve (O1)–(O3): if they do not, H_2 has no realization at all. It shows that layer data are needed, since H_2 passes (C) by Proposition 4.1(d).

Proposition 4.4. *The following was computed (Appendix B).*

(a) All 64 choices for the first three steps of H_2 give layered states, and the orders they give are the eight of the proof of Theorem 4.2.
(b) H_1 is rigidly realizable.
(c) H_2 is not realizable with reverse folds.

Moreover H_3 is not realizable with reverse folds, since its third step violates (C) (Proposition 4.1(d) and Lemma 3.3(3)).

Conjecture 4.5 (Three lines)**.** *$\Lambda \notin \mathcal{P}_{\mathrm{rig}}(A)$. That is, no rigidly realizable history of A, with any number of scaffold creases and in any order of steps, has the surviving blocks $\{0, 4\}$, $\{1, 5\}$, $\{3, 7\}$.*

Conjecture 4.6 (No straight surviving partition)**.** *No partition in $\mathcal{P}_{\mathrm{rig}}(A)$ has all its blocks straight.*

Conjecture 4.7 (Three lines, with reverse folds)**.** *$\Lambda \notin \mathcal{P}_{\mathrm{rev}}(A)$.*

Remark 4.8.

(1) *Relation.* Conjecture 4.6 implies Conjecture 4.5, and so does Conjecture 4.7, since $\mathcal{P}_{\mathrm{rig}}(A) \subseteq \mathcal{P}_{\mathrm{rev}}(A)$ (Lemma 3.3(2)). The partitions of A into straight blocks are the five partitions of the set of its three lines: Λ, the three partitions merging two lines, and $\{A\}$.
(2) *Order is free.* The three conjectures constrain only $\mathcal{P}(H)$, so they cover every order in which the blocks may be finally created, e.g. $\{0, 4\}$ first, in the middle, or last.
(3) *Symmetry.* The isometries $x \mapsto -x$, $x \mapsto 4 - x$ and $x \mapsto x + 4$ preserve A and permute its three lines. The model is equivariant under $\mathrm{Iso}(\mathbb{T})$ (the normalisation in Definition 1.2 only changes images by a global isometry), so all three conjectures are invariant under these symmetries.

(4) *Evidence.* H_2 and H_3, the two histories with $\mathcal{P} = \Lambda$ exhibited here, are not rigidly realizable: H_2 by Theorem 4.2, and H_3 because its third step violates (C), hence (F) by Lemma 2.3. Neither is realizable with reverse folds (Proposition 4.4). By contrast H_1, which is rigidly realizable, has only one straight block.

(5) *Status.* Histories with a given surviving partition are unbounded in length, so no conjecture here is a finite check. A counterexample, if one exists, can be found by enumerating combinatorial history types.

(6) *Rigidity is essential.* Suppose instead that creases persist once made and fold angles vary continuously, so that unfolding a crease to angle π does not erase it. Then Λ is realized trivially: fold $\{1,5\}$, fold $\{3,7\}$, unfold both to the flat sheet, fold $\{0,4\}$, and refold the existing creases as $\{1,7\}$ and $\{3,5\}$, as in H_1. Each crease of A is created exactly once, in the blocks $\{1,5\}$, $\{3,7\}$, $\{0,4\}$. More generally, connectedness results for single-vertex configuration spaces (Streinu–Whiteley, *Single-vertex origami and spherical expansive motions*, JCDCG 2004, LNCS 3742; exact hypotheses not re-checked) suggest that any folded state is then reachable.

Appendix A. Verification script for Theorem 4.2

Unit arcs $(a, a+1)$ of the image carry stacks of faces (bottom to top). A fold through the image point c moves the arcs on one side, reflects them via $a \mapsto 2c - a - 1$, reverses each moved stack, and places it on top (over) or at the bottom (under) of the stack at the landing arc. The fold point of each step is the current image of a crease point of that step. In `choices`, `side` selects the labeling (which side moves) and `over` selects $H^{\uparrow}$: the moving side is $H^{\uparrow}$ iff `over`, matching rule (iii).

```
import itertools
def run(choices):
    g = {k: (1, 0) for k in range(1, 9)}        # image map of f_k: x -> e*x+t (mod 8)
    stack = {a: [a+1] for a in range(8)}        # unit arc (a,a+1) -> faces, bottom->top
    img = lambda k, x: (g[k][0]*x + g[k][1]) % 8
    for (pt, face), (side, over) in zip([(1,1), (3,3), (0,1)], choices):
        c = img(face, pt)                       # fold line through image of a crease point
        plus = lambda a: ((a - c) % 8) < 4
        mov = {a for a in range(8) if plus(a) == (side == 0)}
        new = {a: list(stack[a]) for a in range(8) if a not in mov}
        for a in mov:
            b = (2*c - a - 1) % 8
            r = list(reversed(stack[a])); new.setdefault(b, [])
            new[b] = new[b] + r if over else r + new[b]
        for a in mov:
            for k in stack[a]:
                e, t = g[k]; g[k] = (-e, (2*c - t) % 8)
        stack = {a: new.get(a, []) for a in range(8)}
    return stack
M = {3, 4, 5, 6}
res = set()
for ch in itertools.product(itertools.product((0,1),(0,1)), repeat=3):
    st = run(ch); arcs = [a for a in st if st[a]]
    assert len(arcs) == 1 and len(st[arcs[0]]) == 8   # all faces share one image arc
    res.add(tuple(st[arcs[0]]))
for s in sorted(res):                               # 8 orders, none admits a free block
    print(s, set(s[:4]) in (M, set(range(1, 9)) - M))
```

The output consists of the eight orders listed in the proof of Theorem 4.2, each with the value `False`.

Appendix B. Implementation of the layered model

The computed claims of Example 3.5 and Propositions 3.6 and 4.4 come from an implementation of Definitions 2.1, 2.2, 3.1 and 3.2 (`src/rigid.py` of the `fold-history` code, checked by `tools/verify_flatvertex.py`). Directions are floating-point numbers compared with a tolerance of 10^{-7}. A layered state stores, for each arc between consecutive image points, the order of the faces over it. A step is tried with both labelings, both choices of $H^{\uparrow}$, and, with senses, every sense assignment of Definition 3.1. (F), (F$'$) and (F$''$) are checked at one point of each arc of $\mathbb{T} \setminus (E_S \cup q^{-1}(\ell))$. The new order is built at one point of each arc of $\mathbb{T} \setminus (E_S \cup \sigma_\ell(E_S) \cup q^{-1}(\ell))$ and moved by g; it extends to a layer order exactly when it agrees across every arc endpoint that is not an image point of S'. Then (O1)–(O3) are checked. A history is realizable when a depth-first search over these choices reaches its last step. As a check of the implementation against the hand computation of Theorem 4.2, it reproduces Proposition 4.4(a).